

THE INTERN DIARIES

Intern Report

Cohort Retention Basics

WEEK 3 | DAILY JOURNAL ENTRY



LEARNING BY DOING, ONE DAY AT A TIME

COHORT RETENTION BASICS (COHORT PURCHASE)

1. What is a cohort?

Answer: A cohort is a group of customers who share a common experience or attribute within a defined time period. In this project, the Online Retail II dataset doesn't provide an actual signup date, so I define the cohort using the customer's first valid purchase month as a signup or acquisition proxy. I then follow that group over subsequent months

2. Why is cohort retention useful?

Answer: Cohort retention is useful because it separates customers based on when they were acquired and then tracks their behavior over time. Instead of blending customers from different acquisition periods into one overall retention percentage, I can compare the trajectories of individual cohorts and identify stronger or weaker retention patterns

3. Difference between cohort retention and simple retention? Why is M0 always 100%?

Answer: Simple retention gives an overall retention measure, while cohort retention organizes customers by a common starting period and tracks them over their lifecycle. Cohort retention therefore gives much more context about how different acquisition groups behave.

- M0 is the cohort's starting month. The number of active customers at M0 is therefore the same number used as the cohort denominator, so M0 equals 100% by construction.

4. "Signup month" in this dataset is a proxy, not a measurement — and the data physically cannot tell you the difference?

Answer: Naive assumption before: cohort = signup month, because the task said "by signup month."

- **What we now know:** The file has exactly 8 columns — Invoice, StockCode, Description, Quantity, InvoiceDate, Price, Customer ID, Country. There is no registration / created_at / signup field anywhere. So cohort_month = *first valid purchase month*, an acquisition proxy.
- **Proof:** If a signup column existed it would appear in df.columns; it doesn't.
- **Why it matters:** Claiming a signup date we never observed is the one error that would make the whole matrix indefensible.

5. Retention in transactional data is not a smooth decay curve — a later month can beat an earlier one.

Answer: We assume that: "retention falls monotonically; if $M3 > M1$ something's broken."

- **What we now know:** a customer who skips a month and returns later is *normal* for retail, so the row is allowed to rise.
- **Proof:**
 - Our 2009-12 row goes $100 \rightarrow 35 \rightarrow 33 \rightarrow **43** \rightarrow 38 \dots \rightarrow **50**$ (M11) $\dots \rightarrow 41$ (M23) $\rightarrow 20$ (M24). $M3 (43) > M1 (35)$; M11 (50) is the peak.
 - Across the 19 cohorts mature enough to show M6, average $M3 = 21.41\%$ > average $M1 = 20.30\%$.
- **Why it matters:** *"a sharp drop after acquisition, then a bumpy plateau,"* not *"continuously declining."* Saying "declining" would be factually wrong on your own numbers.